



CONFIDENTIAL TECHNICAL REPORT

Maintenance & Health Check (MHC) Report

System Health, Calibration & Optical Performance Assessment

MACHINE & CUSTOMER IDENTITY

MHC EVALUATION & ASSESSMENT

CUSTOMER ACCOUNT
Acme PCB Corp

PLANT / FACILITY
Plant 2 - Penang

PRODUCTION LINE
—

ZONE
—

MACHINE MODEL
ESI 5330 Flex

MACHINE NUMBER / SOURCE
WLVIA#3

SERIAL NUMBER
SN-98765

BASELINE DATE
—

LAST MHC DATE
2026-09-02

SERVICE ENGINEER
Sahafiz
Field Service Engineer

INSPECTION DATE
2026-09-02

RELEASE STATUS
⌚ PENDING REVIEW

02 TABLE OF CONTENTS / REPORT INDEX

REPORT INDEX

14 Subsystem Diagnostics & Certification Modules

PAGE 01 CUSTOMER & MACHINE IDENTITY 1 Section

§01 Cover Page

PAGE 03 EXECUTIVE SUMMARY & HOURS 2 Sections

§03 Executive Summary

§04 Laser Hours

PAGE 04 LASER POWER & STABILITY 1 Section

§05 Laser Power

PAGE 05 OPTICAL BEAM & SPOT QUALITY 1 Section

§06 Beam Profile

PAGE 06 FOCUS & POWER OFFSETS 2 Sections

§07 Focus Optimization

§08 Power Offset

PAGE 07 STAGE & SENSOR CALIBRATION 2 Sections

§09 Stage Calibration

§10 AGC / Scanner Calibration

PAGE 08 THERMAL ENVIRONMENT 1 Section

§11 Temperature Monitoring

PAGE 09 PRODUCT PROCESS & VIA QUALITY 1 Section

§12 Product Process & Via Quality

PAGE 10 CERTIFICATION & SIGN-OFF 3 Sections

§13 Findings

§14 Spare Parts / Recommendations

§15 Buyoff & Approvals

● MAINTENANCE & HEALTH CHECK REPORT STRUCTURE

This formal technical report presents authoritative inspection, telemetry, and calibration records collected for this equipment. Subsystems are documented with baseline comparisons and recorded evidence where applicable.

03 EXECUTIVE SUMMARY

MHC EVALUATION

MHC RESULT

ACTION REQUIRED – 33%

ACTION_REQUIRED

Routine Maintenance & Health Check (MHC) completed for Flex Drill WLVI#3 (SN-98765) at Acme PCB Corp - Plant 2 - Penang.

MHC INSPECTION RESULTS

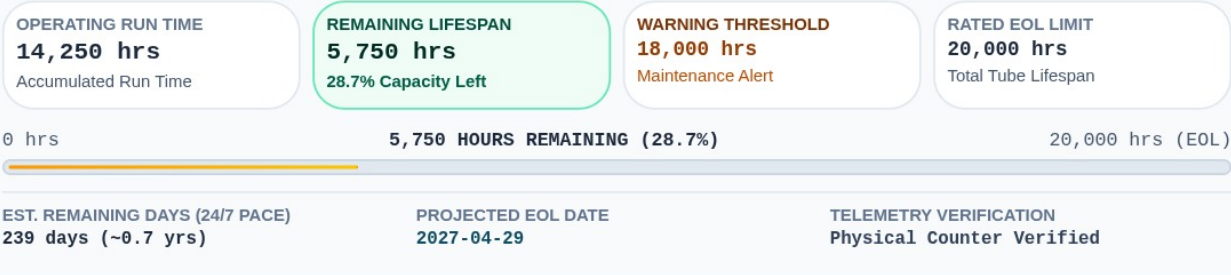
SUBSYSTEM / AUDIT ITEM	SPECIFICATION	VERDICT
Laser Power (Laser Head 1 & 2)	15.0 W ± 10% (13.5–16.5 W)	WARNING
Beam Profile / Mode	–	PASS
Stage Calibration	–	NOT COLLECTED
AGC / Scanner Calibration	–	NOT COLLECTED
Temperature & Cooling	–	PASS

04 LASER LIFECYCLE & USAGE TELEMETRY

1 HEADS ANALYZED

Laser Head 1

PASS



LIFECYCLE PROGNOSIS & SERVICE RECOMMENDATIONS

PROGNOSTIC ADVISORY

Laser Head 1: Mid-to-late life phase (28.7% remaining). Tube capacity is stable. Monitor optical power decay during regular service.

05 LASER POWER & BASELINE COMPARISON

Authoritative Optical Power Measurement & Calibration
Variance Analysis

SPEC: 15.0W ± 10%
(13.50-16.50 W)

COMPLETE

● LEFT: Historical Baseline
(Previous)

VS ● RIGHT: Present Measurement
(Current)

→ ● VARIATION: Δ = Current - Previous

Laser Head 1

SPEC: 15.00 W ± 10% (13.50-16.50 W)

PASS

PREVIOUS (BASELINE)

Baseline

—

Historical Baseline

VS
COMPARE

CURRENT (PRESENT)

2026-09-02

15.10 W

Target: 15.00 W

VARIATION = CURRENT - PREVIOUS

Calibration Shift:

No previous baseline

Laser Head 2

SPEC: 15.00 W ± 10% (13.50-16.50 W)

WARNING

PREVIOUS (BASELINE)

Baseline

—

Historical Baseline

VS
COMPARE

CURRENT (PRESENT)

2026-09-02

Not Recorded

Target: 15.00 W

VARIATION = CURRENT - PREVIOUS

Calibration Shift:

No previous baseline

06 OPTICAL BEAM PROFILE & SPOT QUALITY

Spatial Beam Distribution & Multistage Aperture Quality Telemetry

PASS

LASER HEAD 1

Laser Head 1

Baseline: 3.500 mm | Δ Variation: +0.000 mm (0.0%)

PASS

1. LASER SOURCE (6A)

PASS

CURRENT MEASURED SPOT SIZE

Ø 3.520 mm

APPLICABLE SPECIFICATION:
3.5 mm ±10% (3.15–3.85 mm)

2. FLAT TOP (6B)

PASS

CURRENT MEASURED SPOT SIZE

Ø 4.150 mm

APPLICABLE SPECIFICATION:
4.2 mm ±5% (3.99–4.41 mm)

APERTURE MASK PROFILE EVIDENCE (0.9 MM – 2.2 MM)

6 / 6 Within Target Spec

0.9mm

PASS

Ø 0.950 mm
≥0.9mm

1.1mm

PASS

Ø 1.150 mm
≥1.1mm

1.3mm

PASS

Ø 1.350 mm
≥1.3mm

1.8mm

PASS

Ø 1.850 mm
≥1.8mm

2.0mm

PASS

Ø 2.050 mm
≥2.0mm

2.2mm

PASS

Ø 2.250 mm
≥2.2mm

LASER HEAD 2

Laser Head 2

Baseline: 3.500 mm | Δ Variation: +0.000 mm (0.0%)

PASS

1. LASER SOURCE (7A)

PASS

CURRENT MEASURED SPOT SIZE

Ø 3.480 mm

APPLICABLE SPECIFICATION:
3.5 mm ±10% (3.15–3.85 mm)

2. FLAT TOP (7B)

PASS

CURRENT MEASURED SPOT SIZE

Ø 4.150 mm

APPLICABLE SPECIFICATION:
4.2 mm ±5% (3.99–4.41 mm)

APERTURE MASK PROFILE EVIDENCE (0.9 MM – 2.2 MM)

6 / 6 Within Target Spec

0.9mm

PASS

Ø 0.950 mm
≥0.9mm

1.1mm

PASS

Ø 1.150 mm
≥1.1mm

1.3mm

PASS

Ø 1.350 mm
≥1.3mm

1.8mm

PASS

Ø 1.850 mm
≥1.8mm

2.0mm

PASS

Ø 2.050 mm
≥2.0mm

2.2mm

PASS

Ø 2.250 mm
≥2.2mm

07 FOCUS OPTIMIZATION

Machining Focus Calibration & Focal Sequence Verification

FOCUS ADJUSTMENT DATE
2026-09-02

ADJUSTMENT REASON
Laser source replacement

BASELINE
-0.300 mm

EVALUATION
Laser source optical alignment verified via dummy wafer drill matrix. Focus set to 0.00 position.

LASER HEAD 1

BASELINE: -0.300 mm

+3

Laser Head 1 Wafer Drill

+0.300 mm
Focus Position

+2

Laser Head 1 Wafer Drill

+0.200 mm
Focus Position

+1

Laser Head 1 Wafer Drill

+0.100 mm
Focus Position

0

Laser Head 1 Wafer Drill

0.000 mm
Focus Position

-1

Laser Head 1 Wafer Drill

-0.100 mm
Focus Position

-2

Laser Head 1 Wafer Drill

-0.200 mm
Focus Position

-3

Laser Head 1 Wafer Drill

-0.300 mm
Focus Position

LASER HEAD 2

BASELINE: -0.300 mm

+3

Laser Head 2 Wafer Drill

+0.300 mm
Focus Position

+2

Laser Head 2 Wafer Drill

+0.200 mm
Focus Position

+1

Laser Head 2 Wafer Drill

+0.100 mm
Focus Position

0

Laser Head 2 Wafer Drill

0.000 mm
Focus Position

-1

Laser Head 2 Wafer Drill

-0.100 mm
Focus Position

-2

Laser Head 2 Wafer Drill

-0.200 mm
Focus Position

-3

Laser Head 2 Wafer Drill

-0.300 mm
Focus Position

Note: Top via impact: Focus adjustment primarily affects top diameter (~90%), significantly influencing top via.

08 POWER OFFSET & CALIBRATION

Power Offset & Process Optimization

Power Offset Range: -20% to +20%

LASER HEAD 1 (LH1)

POWER OFFSET: —

PHASE 1

Recipe: —

Adjusted: —

PHASE 2

Recipe: —

Adjusted: —

OFFSET COMPARISON
ADJUSTMENT REASON
—

Prev: — → Curr: —

LASER HEAD 2 (LH2)

POWER OFFSET: —

PHASE 1

Recipe: —

Adjusted: —

PHASE 2

Recipe: —

Adjusted: —

OFFSET COMPARISON
ADJUSTMENT REASON
—

Prev: — → Curr: —

Note: Bottom via impact: Power offset primarily influences bottom via diameter (~90%), making it a key factor in bottom via diameter control.

09 STAGE CALIBRATION (X/Y DEVIATION)

Sub-Micron Motion Stage Positional Accuracy & Cross-Axis Verification

TOLERANCE: ± 2.0
 μm

— NOT
COLLECTED

STAGE IDENTIFIER	X DEVIATION RANGE [MEASURED]	Y DEVIATION RANGE [MEASURED]	MAX ABS DEV [DERIVED]	VERDICT
Stage 1 (Left)	—	—	—	— UNANSWERED
Stage 2 (Right)	—	—	—	— UNANSWERED
Calibration Record: Stage positional calibration completed across the full travel range, with X/Y deviation verified against the defined tolerance.				

10 AGC / SCANNER CALIBRATION

Galvo Scanner & Automatic Grid Calibration Positional Verification

TOLERANCE: ± 3.0
 μm

— NOT
COLLECTED

AGC IDENTIFIER	X DEVIATION RANGE [MEASURED]	Y DEVIATION RANGE [MEASURED]	VERDICT
AGC 1	—	—	— UNANSWERED
AGC 2	—	—	— UNANSWERED
Calibration Record: AGC/scanner positional calibration completed across the full travel range, with X/Y deviation verified against the defined tolerance.			

11 TEMPERATURE & THERMAL TELEMETRY

Optics Markbox 6-Channel Air-Cooling & Thermal Telemetry Monitoring

COMPLETE

OPTICS MARKBOX AIR-COOLING SUBSYSTEM OVERVIEW

TARGET SPEC: —

MARKBOX 1NO DATA

Not Linked

Optics Markbox 1 (CH1 + CH4)

MARKBOX 2NO DATA

Not Linked

Optics Markbox 2 (CH2 + CH5)

MARKBOX 3NO DATA

Not Linked

Optics Markbox 3 (CH3 + CH6)

NO CONTINUOUS MULTI-CHANNEL THERMAL LOG LINKED TO THIS SESSION

To populate detailed 6-channel optics markbox sensor matrices and profile curves,
link a temperature log record to this machine.

ENGINEER OBSERVATION: Cooling optimal

12 PRODUCT PROCESS & VIA QUALITY

Product Recipe Parameters & Via Measurements

✓ COMPLETE

PRODUCT

Flex Rigid Standard

RECIPE / PROGRAM

REC-01

LOT / PANEL IDENTIFIER

SMP-100

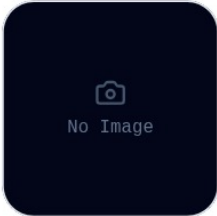
PROCESS RECIPE PHASE PARAMETERS

PROCESS PHASE	POWER (W)	FREQUENCY (kHz)	SHOT COUNT	MASK (mm)	DEFOCUS (mm)
Phase 1	-	-	-	-	-
Phase 2	-	-	-	-	-

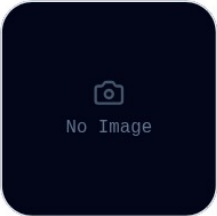
VIA DRILLING MEASUREMENTS & SPECIFICATION

LASER HEAD	TOP DIA.	BOT DIA.	TAPER	VERDICT
Acceptance Spec	-	-	-	Engineering Spec
Laser Head 1 (LH1)	50.0 μm	-	-	✓ PASS
Laser Head 2 (LH2)	-	-	-	✓ PASS

VIA CROSS-SECTION EVIDENCE



No Image



No Image

Laser Head 1 (LH1)Laser Head 2 (LH2)

Engineer Remarks: Profile OK

13 FINDINGS & OBSERVATIONS

TOTAL RECORDED: 0

— NOT COLLECTED

—

ALL SUBSYSTEMS NOMINAL

14 SPARE PARTS & RECOMMENDATIONS

✓ COMPLETE

ENGINEERING RECOMMENDATIONS & FUTURE ACTION PLAN

—

CONSUMED PARTS (SERVICE EXECUTION)

PART NAME	QTY	ACTION	COSTING
Air Filter Element	1	REPLACED	CUSTOMER_COST

RECOMMENDED SPARE PARTS (PROCUREMENT / STOCK)

—

15 BUYOFF & OFFICIAL APPROVALS

PRODUCTION RELEASE: PENDING

🕒 PENDING REVIEW

NEXT SCHEDULED MHC

Target Due Date: 2026-12-01

Cycle: 90-Day Standard Interval

FIELD SERVICE ENGINEER

✓ VERIFIED

Sahafiz

Field Service Engineer
Date: 2026-09-02

[COMPLETED & SIGNED BY ENGINEER]

CUSTOMER ACCEPTANCE REPRESENTATIVE

🟡 PENDING

Pending Customer Sign-off

Acme PCB Corp
Date: —

[PENDING CUSTOMER REVIEW & SIGN-OFF]